

THE FUG INPUTS (CASE DATA)

CH₄ / C₂H₄ splitter shortcut-validation basis

Process basis

Item	Specified case value
Column configuration	Total condenser; kettle reboiler
Operating pressure	7.0 bar
Feed rate	24,000 kg/h
Feed mass fractions	CH ₄ = 0.62; C ₂ H ₄ = 0.38
Feed thermal condition	Subcooled; use $q = 1.2$
Relative volatility	Varies with location in the column (see note below)
Key assignment	Light key = methane (CH ₄); heavy key = ethene (C ₂ H ₄)

Relative volatility. Vapor-liquid equilibrium checks at the two ends of the column give $\alpha(\text{CH}_4/\text{C}_2\text{H}_4) = 2.500$ at the distillate-end (dew-point) conditions and $\alpha(\text{CH}_4/\text{C}_2\text{H}_4) = 1.369$ at the bottoms-end (bubble-point, kettle reboiler) conditions. Relative volatility varies with temperature and composition across the column; select a single representative value appropriate for the shortcut Fenske-Underwood-Gilliland method.

Product-composition basis

Overhead GC interpretation. The top GC tag was mislabelled in the original design report, but the analyser is correctly installed on the vapor line above the total condenser and reports vapor peak ratios. Normalize the reported vapor ratio directly and use the resulting composition as the distillate composition.

Measurement	Specified value
Overhead GC vapor peak ratio	CH ₄ : C ₂ H ₄ = 18.0 : 1.00
Bottoms composition	$x_{\text{B,CH}_4} = 0.0100$; $x_{\text{B,C}_2\text{H}_4} = 0.9900$

Gilliland / Eduljee operating-reflux basis

Use an operating reflux ratio of $R = 1.25 \times R_{\text{min}}$. For the Gilliland stage estimate, use the Eduljee explicit form:

$X = (R - R_{\text{min}}) / (R + 1)$
$Y = 0.75 \times (1 - X^{0.5668})$

$$N = (N_{\min} + Y) / (1 - Y)$$

Use full precision in intermediate calculations. Apply the reporting precision stated in the task prompt only to final reported values.